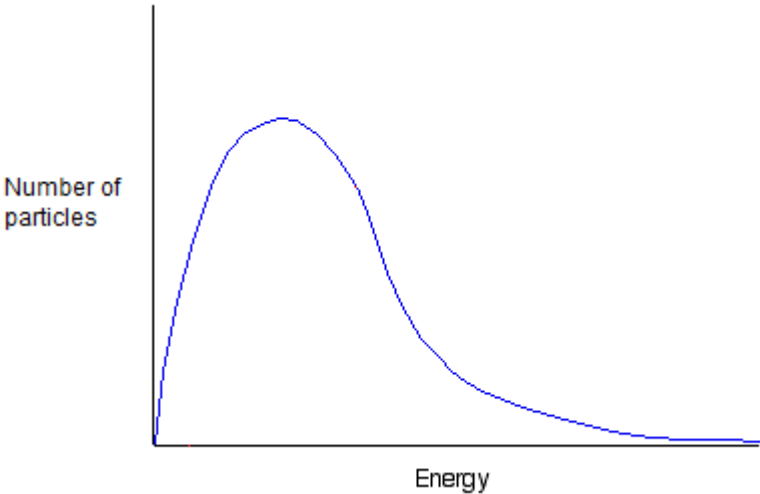


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		 axes labelled (1) curve – start at origin, hump, does not touch x-axis (1)	2			2		
		(ii)		activation energy E_a is lower with a catalyst / E_a with catalyst and without catalyst marked on diagram (1) more particles / collisions exceed E_a with catalyst (1)		2		2		
				Question 12 total	2	7	4	13	6	6